A COUNTABLY GENERATED SUBMODULE OF A BASE CHANGE IS CONTAINED IN THE IMAGE OF A COUNTABLY GENERATED SUBMODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Expansion of a tensor as a finite sum of pure tensors. Used in the counted proof at lines 22649-22650. Produces the coefficients whose generated submodule is the candidate P.
- Countability of a countable union of finite sets. Implicit in the construction of P at line 22650. Keeps the generating set of P countable, which is what the statement requires.

Lemma 0.1. *Let $R \to S$ be a ring map, let M be an R-module, and let Q be a countably generated S-submodule of $M \otimes_R S$. Then there exists a countably generated R-submodule P of M such that* $\mathrm{Im}(P \otimes_R S \to M \otimes_R S)$ *contains Q.*

Proof. Let $y_1, y_2, \ldots$ be generators for Q and write $y_j = \sum_k x_{jk} \otimes s_{jk}$ for some $x_{jk} \in M$ and $s_{jk} \in S$. Then take P be the submodule of M generated by the x_{jk}. □

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